

Recovering the unwritten recurrences of the table OEIS A251801

Adrian Perez Fontelles

Independent researcher

2 September 2026

Abstract

OEIS A251801 is a two-dimensional table $T(n, k)$ whose empirical recurrences are, for several of its lines, given only as an ORDER — “k=2: [order 20]” and the like — with the coefficients in a linked file rather than in the entry. Those conjectures can still be settled, and the recurrences recovered. A line of the table is a fixed-width or fixed-height array count, hence a walk count on finitely many vertices, hence satisfies some monic recurrence of order at most that number; Berlekamp–Massey on twice as many exact terms returns the MINIMAL one. Where its order equals the order the entry states — and where the same holds after the first few terms are dropped — any recurrence of that order the line satisfies has a characteristic polynomial that is a multiple of the minimal one and of equal degree, hence is that polynomial. This note settles column 2, row 2, row 3.

2020 Mathematics Subject Classification. 05A15, 11B37, 15A18.

1 The table and the unwritten conjectures

OEIS A251801 is “T(n,k)=Number of (n+1)X(k+1) 0..3 arrays with no 2X2 subblock having x11-x00 less than x10-x01.”. It is read by antidiagonals and begins

150, 1080, 1740, 6627, 28932, 19269, 36552, 348679, ...

The entry carries, among its empirical claims:

k=2: [order 20]
n=2: [order 22]
n=3: [order 40]

As of the “Last modified” line on the live entry (Jun 02 2025 10:32:19, revision 6) these are still recorded as empirical, and the recurrences themselves are not in the entry text.

2 Why a line of the table is a walk count

Fix the index that the line fixes. The entry defines $T(n, k)$ as a number of arrays of a shape determined by n and k , subject to a condition local to a bounded window of consecutive lines. Substituting the fixed index into the entry’s own wording turns the definition into an ordinary one-parameter array count, and for such a count the admissible windows are the vertices of a finite digraph and an array is a walk. So the line is a walk count on S vertices, and by the Cayley–Hamilton theorem it satisfies a monic integer recurrence of order at most S .

3 Recovering the recurrences

Lemma 1. *A sequence known to satisfy some linear recurrence of order at most S is determined, with its minimal recurrence, by its first $2S$ terms; Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Column $k = 2$

Substituting the fixed index into the entry's wording gives an ordinary one-parameter array count, a walk on $S = 64$ vertices. Berlekamp–Massey on $2S = 128$ exact terms returns a minimal recurrence of order 20 — the order the entry states — with integer coefficients

c_1	30	c_6	−170647	c_{11}	2262184	c_{16}	−12352
c_2	−40	c_7	−169762	c_{12}	3737187	c_{17}	−613952
c_3	−2202	c_8	1201353	c_{13}	−3793360	c_{18}	105792
c_4	7167	c_9	−237196	c_{14}	−1475516	c_{19}	54144
c_5	41092	c_{10}	−3305290	c_{15}	2385944	c_{20}	−13824

so that $a(m) = \sum_{i=1}^{20} c_i a(m-i)$ for every m . Removing the first $20 + 4$ terms and repeating gives order 20 again, which is what the identification below needs.

Row $n = 2$

Substituting the fixed index into the entry's wording gives an ordinary one-parameter array count, a walk on $S = 64$ vertices. Berlekamp–Massey on $2S = 128$ exact terms returns a minimal recurrence of order 22 — the order the entry states — with integer coefficients

c_1	40	c_7	8833486	c_{13}	492639700	c_{19}	20037984
c_2	−755	c_8	−27833039	c_{14}	−498877265	c_{20}	−4463424
c_3	8946	c_9	71958408	c_{15}	414825042	c_{21}	625536
c_4	−74677	c_{10}	−153781873	c_{16}	−280100140	c_{22}	−41472
c_5	467156	c_{11}	272810702	c_{17}	151065320		
c_6	−2274363	c_{12}	−402324879	c_{18}	−63500432		

so that $a(m) = \sum_{i=1}^{22} c_i a(m-i)$ for every m . Removing the first $22 + 4$ terms and repeating gives order 22 again, which is what the identification below needs.

Row $n = 3$

Substituting the fixed index into the entry's wording gives an ordinary one-parameter array count, a walk on $S = 256$ vertices. Berlekamp–Massey on $2S = 512$ exact terms returns a minimal recurrence of order 40 — the order the entry states — with integer coefficients

$$(c_1, \dots, c_{40}) = (70, -2375, 52042, -827990, 10195132, -101124322, 830421108, -5758451859, 34223487178,$$

so that $a(m) = \sum_{i=1}^{40} c_i a(m-i)$ for every m . Removing the first $40 + 4$ terms and repeating gives order 40 again, which is what the identification below needs.

Theorem 2. *For each line above, the displayed recurrence holds for every index, and it is the recurrence the entry records.*

Proof. It holds by Lemma 1. For the identification, the entry asserts that the line satisfies SOME recurrence of the stated order, possibly only from some index on. Let q be its characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail. Then $q_{\min} \mid q$, and the second Berlekamp–Massey run — on the line with its first few terms removed — shows $\deg q_{\min}$ equals the stated order, which is $\deg q$. Both are monic, so $q = q_{\min}$. $\square$

4 Verification

Three checks.

First, each line's model was compared against the entry's own published values for that line, taken from the antidiagonal DATA read in either orientation and from the "Table starts" block, and agrees at every available term. This is the only point at which reading the entry's English enters, and a misreading gives different counts.

Second, each recovered recurrence was evaluated directly on the model's values, with no Berlekamp–Massey involved, and holds at every index tested.

Third, each order was computed twice by independent routes — modulo a 61-bit prime, where every intermediate is a machine word, and again in exact rational arithmetic — and once more on the line with its first few terms dropped, which is what the identification requires. All agree.

References

- [1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, <https://oeis.org>, 2026. Sequence A251801.
- [2] J. L. Massey, Shift-register synthesis and BCH decoding, *IEEE Trans. Inform. Theory* **15** (1969), 122–127.
- [3] R. P. Stanley, *Enumerative Combinatorics*, Volume 1, 2nd ed., Cambridge University Press, 2011. (Transfer-matrix method, Section 4.7.)
- [4] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.